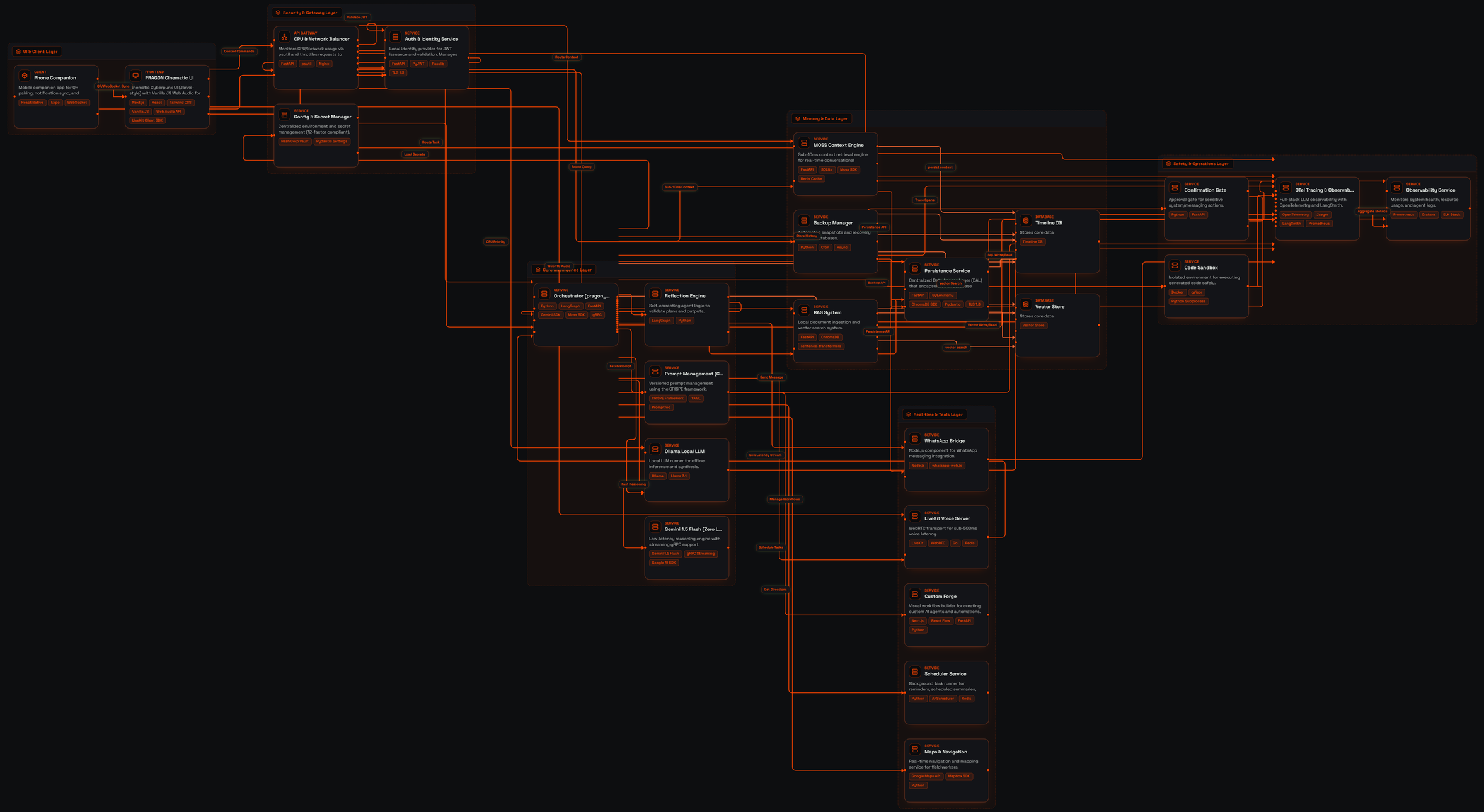
UI & Client Layer
CLIENT
Phone Companion
Mobile companion app for QR pairing, notification sync, and
React Native
Expo
WebSocket
QR/WebSocket Sync
FRONTEND
PRAGON Cinematic UI
inematic Cyberpunk UI (Jarvis-style) with Vanilla JS Web Audio for
Next.js
React
Tailwind CSS
Vanilla JS
Web Audio API
LiveKit Client SDK
Control Commands
Security & Gateway Layer
Validate JWT
API GATEWAY
CPU & Network Balancer
Monitors CPU/Network usage via psutil and throttles requests to
FastAPI
psutil
Nginx
SERVICE
Auth & Identity Service
Local identity provider for JWT issuance and validation. Manages
FastAPI
PyJWT
Passlib
TLS 1.3
Route Context
SERVICE
Config & Secret Manager
Centralized environment and secret management (12-factor compliant).
HashiCorp Vault
Pydantic Settings
Route Task
Load Secrets
Route Query
Memory & Data Layer
SERVICE
MOSS Context Engine
Sub-10ms context retrieval engine for real-time conversational
FastAPI
SQLite
Moss SDK
Redis Cache
Sub-10ms Context
SERVICE
Backup Manager
Automated snapshots and recovery databases.
Store History
Persistence API
Python
Cron
Rsync
CPU Priority
WebRTC Audio
Core Orchestration Layer
SERVICE
Orchestrator (pragon_...
Python
LangGraph
FastAPI
Gemini SDK
Moss SDK
gRPC
SERVICE
Reflection Engine
Self-correcting agent logic to validate plans and outputs.
LangGraph
Python
SERVICE
RAG System
Local document ingestion and vector search system.
FastAPI
ChromaDB
sentence-transformers
Persistence API
Backup API
SERVICE
Persistence Service
Centralized Data Access Layer (DAL) that encapsulates database
FastAPI
SQLAlchemy
ChromaDB SDK
Pydantic
TLS 1.3
Vector Search
Vector Write/Read
vector search
SQL Write/Read
persist context
Trace Spans
DATABASE
Timeline DB
Stores core data
Timeline DB
DATABASE
Vector Store
Stores core data
Vector Store
Safety & Operations Layer
SERVICE
Confirmation Gate
Approval gate for sensitive system/messaging actions.
Python
FastAPI
SERVICE
Code Sandbox
Isolated environment for executing generated code safely.
Docker
gVisor
Python Subprocess
SERVICE
OTel Tracing & Observab...
Full-stack LLM observability with OpenTelemetry and LangSmith.
OpenTelemetry
Jaeger
LangSmith
Prometheus
Aggregate Metrics
SERVICE
Observability Service
Monitors system health, resource usage, and agent logs.
Prometheus
Grafana
ELK Stack
Fetch Prompt
SERVICE
Prompt Management (C...
Versioned prompt management using the CRISPE framework.
CRISPE Framework
YAML
Promptfoo
Send Message
SERVICE
Ollama Local LLM
Local LLM runner for offline inference and synthesis.
Ollama
Llama 3.1
Low Latency Stream
Fast Reasoning
Manage Workflows
SERVICE
Gemini 1.5 Flash (Zero L...
Low-latency reasoning engine with streaming gRPC support.
Gemini 1.5 Flash
gRPC Streaming
Google AI SDK
Schedule Tasks
Get Directions
Real-time & Tools Layer
SERVICE
WhatsApp Bridge
Node.js component for WhatsApp messaging integration.
Node.js
whatsapp-web.js
SERVICE
LiveKit Voice Server
WebRTC transport for sub-500ms voice latency.
LiveKit
WebRTC
Go
Redis
SERVICE
Custom Forge
Visual workflow builder for creating custom AI agents and automations.
Next.js
React Flow
FastAPI
Python
SERVICE
Scheduler Service
Background task runner for reminders, scheduled summaries,
Python
APScheduler
Redis
SERVICE
Maps & Navigation
Real-time navigation and mapping service for field workers.
Google Maps API
Mapbox SDK
Python